Non-Dilutive Open-Source Public-Good Grant Dossier (‘pqc-immune-daemon

Target Grant Programs: NLnet Foundation (NGI Zero Core / Entrust), OpenSSF, Sovereign Tech Fellowships
Applicant Structure: Independent Individual Systems & Cryptography Researcher (Non-Corporate, Zero-Equity, Open-Source Public Good)
Public Repository: ‘pqc-immune-daemon‘ (Apache-2.0 / MIT Dual License)
Requested Grant Range: ‘?15,000 ? ?50,000‘ (Milestone-Based Individual Fellowship)

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1. Abstract & Public-Good Mission

Edge Linux nodes, autonomous software agents, and decentralized infrastructure increasingly face two converging threats:
1. **Harvest-Now, Decrypt-Later Quantum Adversaries** targeting legacy RSA/ECDH transport channels.
2. **Time-of-Check to Time-of-Use (TOCTOU) Runtime Binary Substitution & Policy Bypass** (such as kernel/userspace descriptor races).

‘pqc-immune-daemon‘ provides a lightweight, fail-closed post-quantum edge security membrane combining:
- **Post-Quantum Hybrid Key Encapsulation:** NIST FIPS 203 (‘ML-KEM-768‘) / ‘FrodoKEM-976-AES‘ with ‘X25519‘ hybrid failover and FIPS 204 (‘ML-DSA-65‘) / ‘Ed25519‘ detached provenance signatures.
- **Formally Verified Policy & Decision Core (‘QF-A3‘ / ‘QF-A4‘):** Machine-checked F* specifications extracted to deterministic C11 via KaRaMeL and paired with an in-process WebAssembly Open Policy Agent (‘OPA Wasm‘) gatekeeper.
- **Descriptor-Bound Runtime Execution (‘QF-A6‘):** Eliminates binary substitution race conditions between cryptographic verification and execution by enforcing ‘EXECUTED_FD == VERIFIED_FD‘ via Linux ‘fexecve‘ and ‘memfd_create‘.

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2. Measurable Technical Milestones (Pay-Per-Milestone)

Milestone	Public-Good Deliverable	Verification & Acceptance Test	Grant Allocation
M1: Formal Core & C11 Differential Suite	Publish F* 2026.09 proofs and KaRaMeL-extracted C11 admission core with ‘10,000+‘ differential parity vectors.	Automated CI pass (‘verify_formal_core.sh‘)	‘?12,000‘
M2: Descriptor-Bound ‘fexecve‘ Runtime Loader	Release Linux ‘fexecve‘ runtime admission wrapper preventing TOCTOU binary modification on disk.	Adversarial race-condition test suite (‘100%‘ block rate)	‘?12,000‘
M3: In-Process OPA Wasm Policy Membrane	Zero-network-hop Rego-to-Wasm policy evaluator with bit-for-bit Python/C/Wasm decision parity.	‘61/61‘ adversarial & boundary test suite pass	‘?10,000‘
M4: Reproducible CBOM & Independent Verifier	Standalone cryptographic bill of materials (CBOM) and Merkle provenance verifier (‘verify_dossier_independent.py‘).	Reproducible build & independent audit pass	‘?6,000‘

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3. Why This Serves the Greater Good Without Corporate Gravity (‘QF-M5‘ Grant Readiness)

- **100% Free & Open Source Deliverable:** Every line of code, F* proof, and test vector in ‘pqc-immune-daemon‘ is released publicly under Apache-2.0 / MIT so hospitals, municipalities, journalists, and independent developers can harden Linux edge systems at ‘$0.00‘ licensing cost.
- **Reproducible Technical Demonstrations:** Every milestone is backed by deterministic verification scripts (‘verify_formal_core.sh‘, ‘verify_dossier_independent.py‘) and a pinned SHA-256 evidence manifest.
- **Realistic Maintenance & Household Boundary:** Milestones are scoped to self-contained, zero-SLA open-source releases (‘<= 6 hours/day‘ under ‘QF-OPS-01‘) that never compromise Operator household stability.
- **Safeguard A & C Compliance:** All individual grant agreements and tax reporting obligations are honored; unreceived grant milestones (‘pending_grants_usd‘) are strictly excluded from $R_{\text{verified liquid}}$ until settled in liquid cash.
- **Zero Cloud Dependency:** Runs entirely on modest commodity hardware (‘<= 14 GiB RAM‘) without requiring proprietary cloud security subscriptions.